0. Introduction

1. Nature of individual life: Death process
2. Nature of individual life: Childbearing

3. Characteristics of population that arise from the nature of individual's life

4. Traits of individuals defined in the population

5. Synthesis of the interdisciplinary science

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l_x : Probability that a newborn survives to age *x*

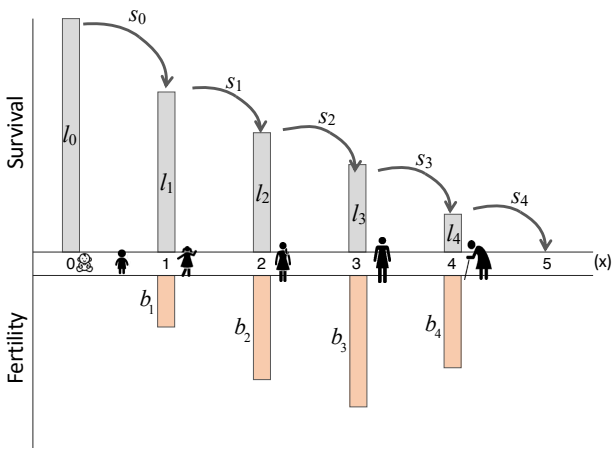
$s_x = \dfrac{l_{x+1}}{l_x}$: Probability that an individual alive at age-*x* survives to age *x* + 1

Age-specific survival rate

b_x : Average number of female offspring produced by an age-*x* female

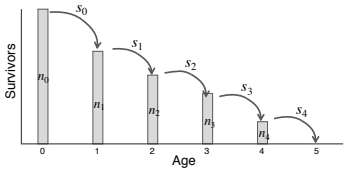
Age-specific fertility rate

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The survival relational equations

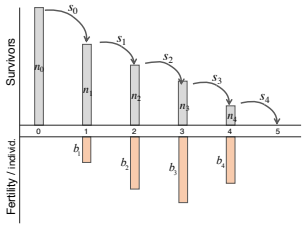


Age	Time t	$t+1$
0	n_0	
1	n_1	n_1
2	n_2	n_2
3	n_3	n_3
4		n_4

$$\left.\begin{aligned} n_1(t+1) &= s_0 n_0(t) \\ n_2(t+1) &= s_1 n_1(t) \\ n_3(t+1) &= s_2 n_2(t) \\ n_4(t+1) &= s_3 n_3(t) \end{aligned}\right| \text{ survival}$$

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The birth and survival relational equations



齢	時間 t	$t+1$
0	n_0	n_0
1	n_1	n_1
2	n_2	n_2
3	n_3	n_3
4		n_4

$$n_0(t+1) = b_1 n_1(t+1) + b_2 n_2(t+1) + b_3 n_3(t+1) + \cdots \mid \text{birth}$$

$$\left.\begin{aligned} n_1(t+1) &= s_0 n_0(t) \\ n_2(t+1) &= s_1 n_1(t) \\ n_3(t+1) &= s_2 n_2(t) \\ n_4(t+1) &= s_3 n_3(t) \end{aligned}\right| \text{ survival}$$

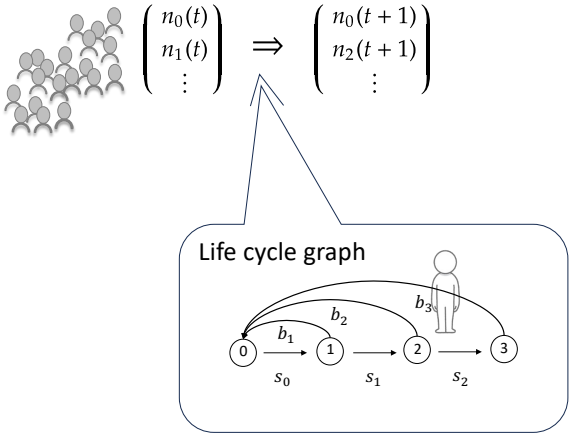
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The birth and survival relational equations

$$n_0(t+1) = b_1 \underbrace{n_1(t+1)}_{n_1(t+1)=s_0 n_0(t)} + b_2 \underbrace{n_2(t+1)}_{n_2(t+1)=s_1 n_1(t)} + \cdots + b_m \underbrace{n_m(t+1)}_{n_m(t+1)=s_{m-1} n_{m-1}(t)}$$

$$\left.\begin{aligned} n_0(t+1) &= b_1 s_0 n_0(t) + b_2 s_1 n_1(t) + \cdots + b_m s_{m-1} n_{m-1}(t) \\ n_1(t+1) &= s_0 n_0(t) \\ n_2(t+1) &= s_1 n_1(t) \\ &\vdots \\ n_m(t+1) &= s_{m-1} n_{m-1}(t) \end{aligned}\right| \begin{array}{l} \text{birth} \\ \\ \\ \\ \text{survival} \end{array}$$

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$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1m}x_m \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2m}x_m \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mm}x_m \end{cases}$$

Math Study

$$\mathbf{Ax} = \underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mm} \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}}_{\mathbf{x}}$$

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$$\begin{cases} n_0(t+1) = b_1 s_0 n_0(t) + b_2 s_1 n_1(t) + \cdots + b_m s_{m-1} n_{m-1}(t) & \text{birth} \\ n_1(t+1) = s_0 n_0(t) \\ n_2(t+1) = s_1 n_1(t) \\ \vdots \\ n_{m-1}(t+1) = s_{m-2} n_{m-1}(t) \end{cases} \quad \text{survival}$$

$$\begin{cases} n_0(t+1) = b_1 s_0 n_0(t) + b_2 s_1 n_1(t) + \cdots + b_m s_{m-1} n_{m-1}(t) \\ n_1(t+1) = s_0 n_0(t) + 0\, n_1(t) + \cdots + 0\ n_{m-1}(t) \\ n_2(t+1) = 0\, n_0(t) + s_1 n_1(t) + \cdots + 0\ n_{m-1}(t) \\ \vdots \\ n_{m-1}(t+1) = 0\, n_0(t) + 0\, n_1(t) + \cdots + 0\ n_{m-1}(t) \end{cases}$$

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Mathematical model of population transition

$$\begin{cases} n_0(t+1) = b_1 s_0 n_0(t) + b_2 s_1 n_1(t) + \cdots + b_m s_{m-1} n_{m-1}(t) \\ n_1(t+1) = s_0 n_0(t) + 0\, n_1(t) + \cdots + 0\ n_{m-1}(t) \\ n_2(t+1) = 0\, n_0(t) + s_1 n_1(t) + \cdots + 0\ n_{m-1}(t) \\ \vdots \\ n_{m-1}(t+1) = 0\, n_0(t) + 0\, n_1(t) + \cdots + 0\ n_{m-1}(t) \end{cases}$$

$$\begin{pmatrix} n_0(t+1) \\ n_1(t+1) \\ \vdots \\ n_{m-1}(t+1) \end{pmatrix} = \begin{pmatrix} s_0 b_1 & s_1 b_2 & \cdots & s_{m-2} b_{m-1} & s_{m-1} b_m \\ s_0 & 0 & \cdots & 0 & 0 \\ 0 & s_1 & \cdots & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & s_{m-2} & 0 \end{pmatrix} \begin{pmatrix} n_0(t) \\ n_1(t) \\ \vdots \\ n_{m-1}(t) \end{pmatrix}$$

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Matrix demography model



Leslie 1942

$$\mathbf{n}(t+1) = \mathbf{L}\,\mathbf{n}(t)$$

$$\mathbf{n}(t+1) = \begin{pmatrix} n_0(t+1) \\ n_1(t+1) \\ \vdots \\ n_{m-1}(t+1) \end{pmatrix}, \quad \mathbf{n}(t) = \begin{pmatrix} n_0(t) \\ n_1(t) \\ \vdots \\ n_{m-1}(t) \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} s_0 b_1 & s_1 b_2 & \cdots & s_{m-2} b_{m-1} & s_{m-1} b_m \\ s_0 & 0 & \cdots & 0 & 0 \\ 0 & s_1 & \cdots & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & s_{m-2} & 0 \end{pmatrix}$$

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